

8. Project Demonstration

Project Title

A Comprehensive Measure of Well-Being using Machine Learning

Demonstration Overview

The demonstration presents the complete workflow of the project, beginning with data collection and preprocessing, followed by model training, evaluation, and deployment through a Flask web application.

Demonstration Steps

Step 1

Launch the Flask application.

Step 2

Open the application in a web browser.

Step 3

Enter the following values:

- * Life Expectancy

- * Mean Years of Schooling

- * Gross National Income Per Capita

Step 4

Click ****Predict HDI****.

Step 5

The application processes the input using the trained Linear Regression model and displays the predicted Human Development Index.

Screenshots to Include

- * Home Page

- * Prediction Form

- * Prediction Result

- * VS Code Project Structure

- * Terminal Showing Flask Running
- * GitHub Repository
- * Model Training Output

Future Enhancements

- * Add multiple machine learning algorithms.
- * Improve prediction accuracy with advanced models.
- * Deploy the application on a cloud platform.
- * Integrate interactive dashboards and visual analytics.
- * Enable user authentication and data history.

Conclusion

The project demonstrates an end-to-end machine learning solution for predicting the Human Development Index. It combines data preprocessing, model development, evaluation, and Flask deployment into a practical and user-friendly application suitable for educational and research purposes.